Calibration results
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Normalized Residuals

Reprojection error (cam0): mean 1.0621151242148579, median 0.9215329801936646, std: 0.7018608575987746
Reprojection error (cam1): mean 1.0145274758672858, median 0.8545509129283693, std: 0.7036384703760202
Gyroscope error (imu0): mean 1.7169387643264098, median 1.4894117988490234, std: 0.9847955168917015
Accelerometer error (imu0): mean 0.44957321394670774, median 0.39361384780831266, std: 0.274934230530444

Residuals

Reprojection error (cam0) [px]: mean 1.0621151242148579, median 0.9215329801936646, std: 0.7018608575987746
Reprojection error (cam1) [px]: mean 1.0145274758672858, median 0.8545509129283693, std: 0.7036384703760202
Gyroscope error (imu0) [rad/s]: mean 0.0171693876432641, median 0.014894117988490232, std: 0.009847955168917014
Accelerometer error (imu0) [m/s^2]: mean 0.17982928557868308, median 0.15744553912332504, std: 0.10997369221217759

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.00218501 0.02784515 0.99960986 0.01191828]
[-0.00266759 0.99960853 -0.02785095 0.16813082]
[-0.99999405 -0.00272741 -0.00210988 -0.1149622]
[0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.00218501 -0.00266759 -0.99999405 -0.11448697]
[0.02784515 0.99960853 -0.00272741 -0.16871042]
[0.99960986 -0.02785095 -0.00210988 -0.00747359]
[0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0002875771984465692

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.01564593 0.02338682 0.99960405 -0.03675999]
[-0.006304   0.99970886 -0.0232906   0.16793131]
[-0.99985772 -0.0059371   0.01578881 -0.11623929]
[0.   0.   0.   1.   ]]

T_ic: (cam1 to imu0):
[[0.01564593 -0.006304   -0.99985772 -0.11458896]
[0.02338682 0.99970886 -0.0059371 -0.16771284]
[0.99960405 -0.0232906   0.01578881 0.04249193]
[0.   0.   0.   1.   ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.00029766560555830106

Baselines:

Baseline (cam0 to cam1):
[[0.99983109 -0.00450399 -0.01781867 -0.04996747]
[0.0045693   0.99998298 0.00362649 0.0001658]
[0.01780203 -0.00370729 0.99983466 -0.00088495]
[0.   0.   0.   1.   ]]
baseline norm: 0.049975578152190045 [m]

Gravity vector in target coords: [m/s^2]
[0.27011565 -9.8028292 -0.00032241]

Calibration configuration
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cam0

Camera model: omni
Focal length: [2044.6753301041977, 2065.5927559749084]
Principal point: [616.3310327581959, 404.1255934969876]
Omni xi: 1.5936757817618359
Distortion model: radtan
Distortion coefficients: [2.290964717370836, -1.8671244246514953, -0.010010836103043635, 0.002831387673779597]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

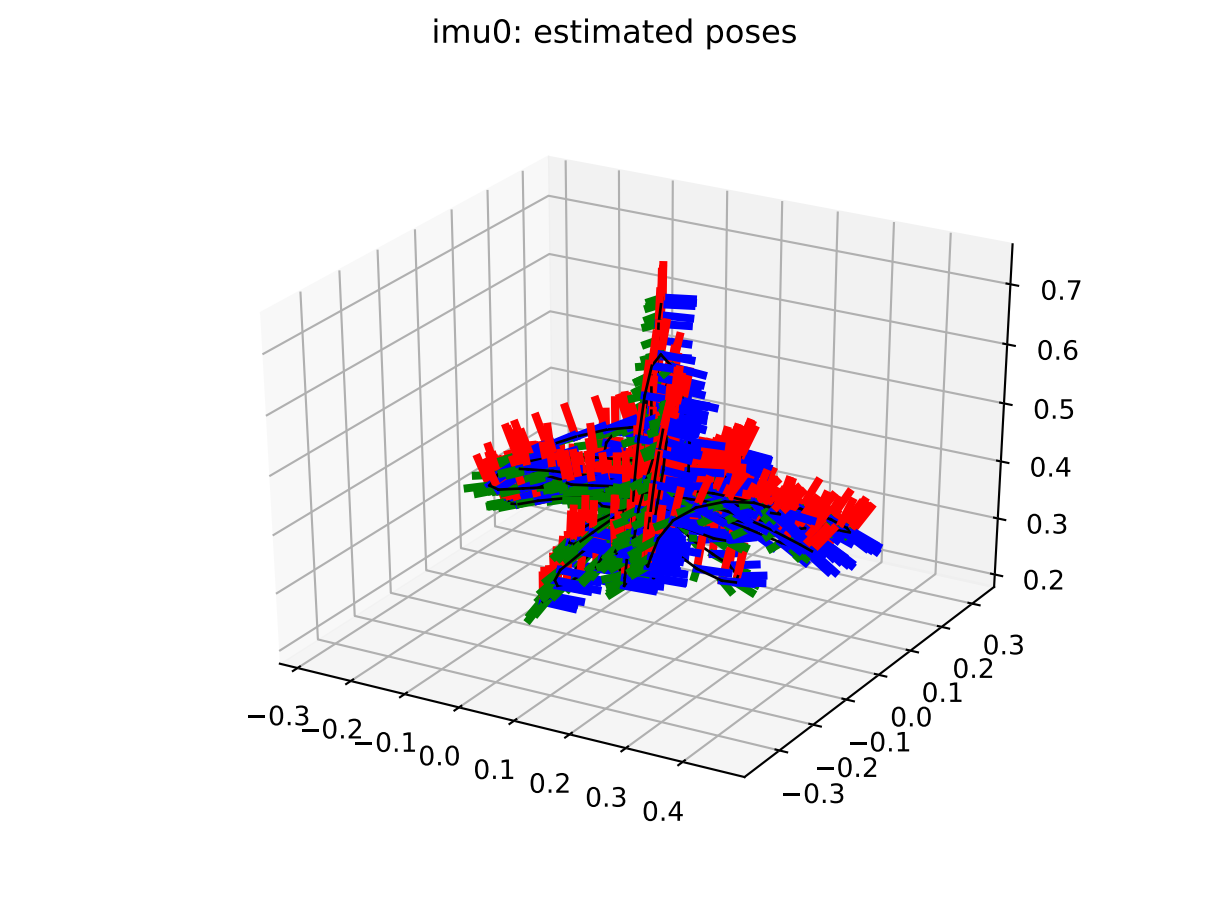
cam1

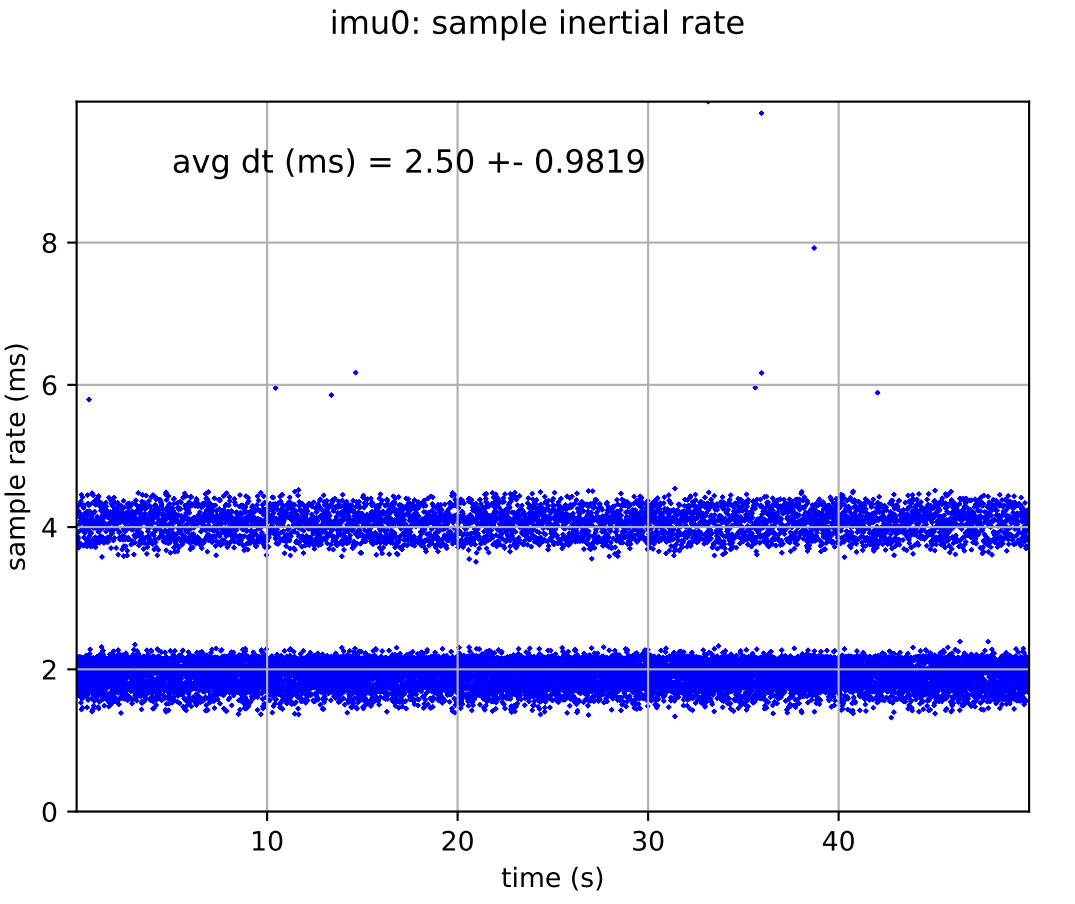
Camera model: omni
Focal length: [2037.2418743359228, 2054.1495623160185]
Principal point: [634.1340582275024, 379.92908128448573]
Omni xi: 1.569995877298825
Distortion model: radtan
Distortion coefficients: [2.167702619581351, -0.9560356055374504, -0.01088770643867036, 0.012396986450845738]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

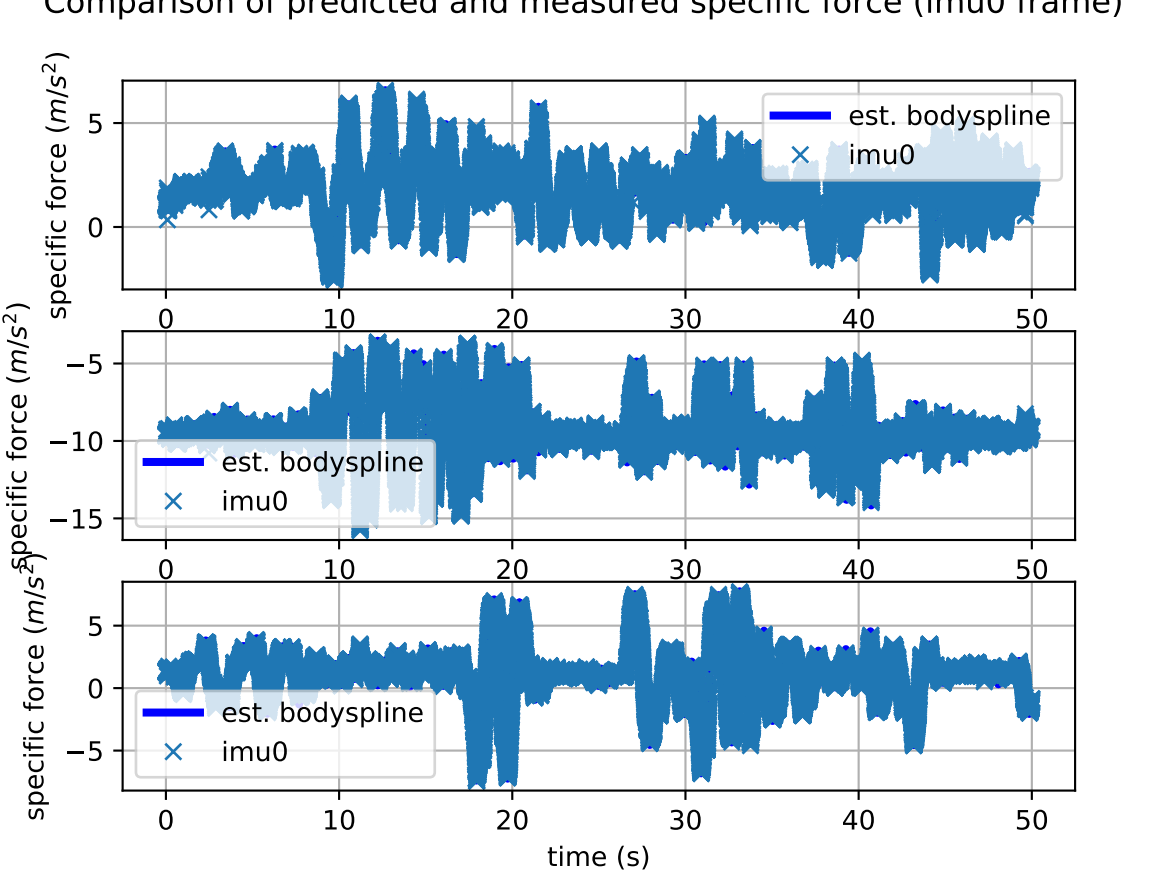
IMU configuration
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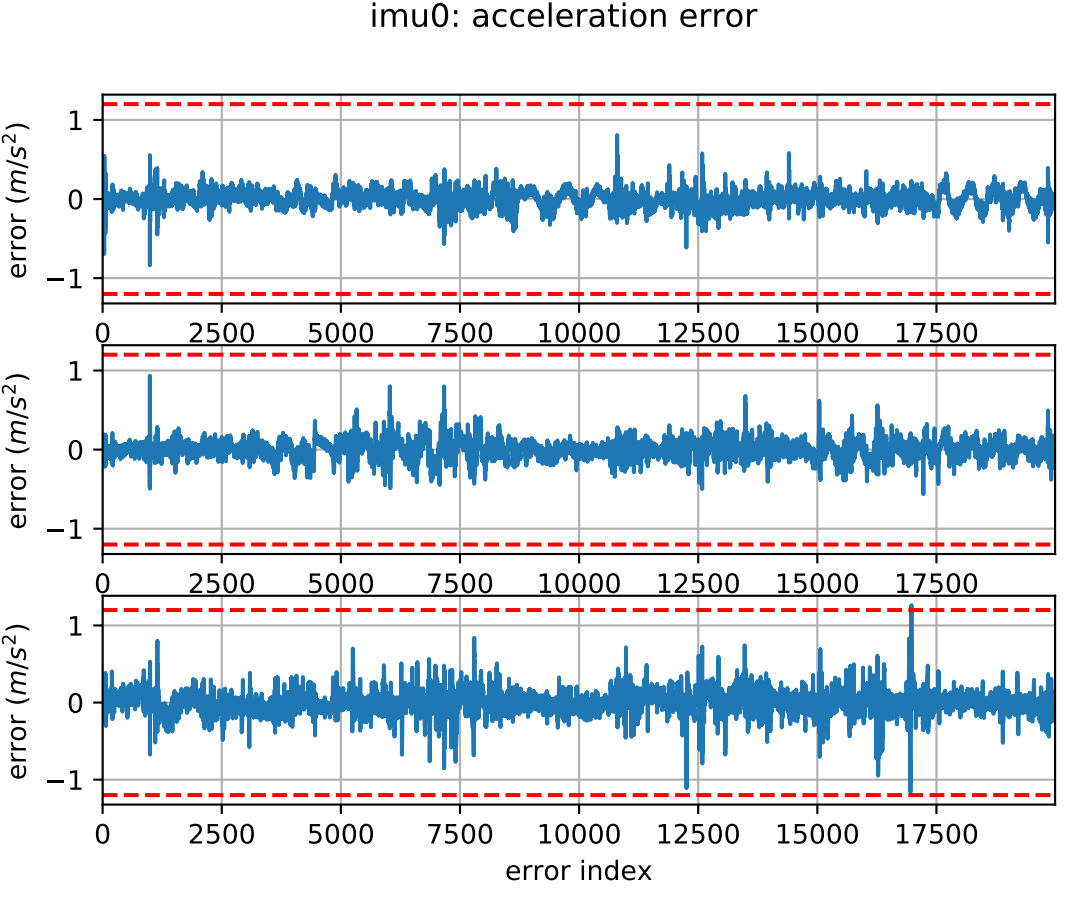
IMU0:

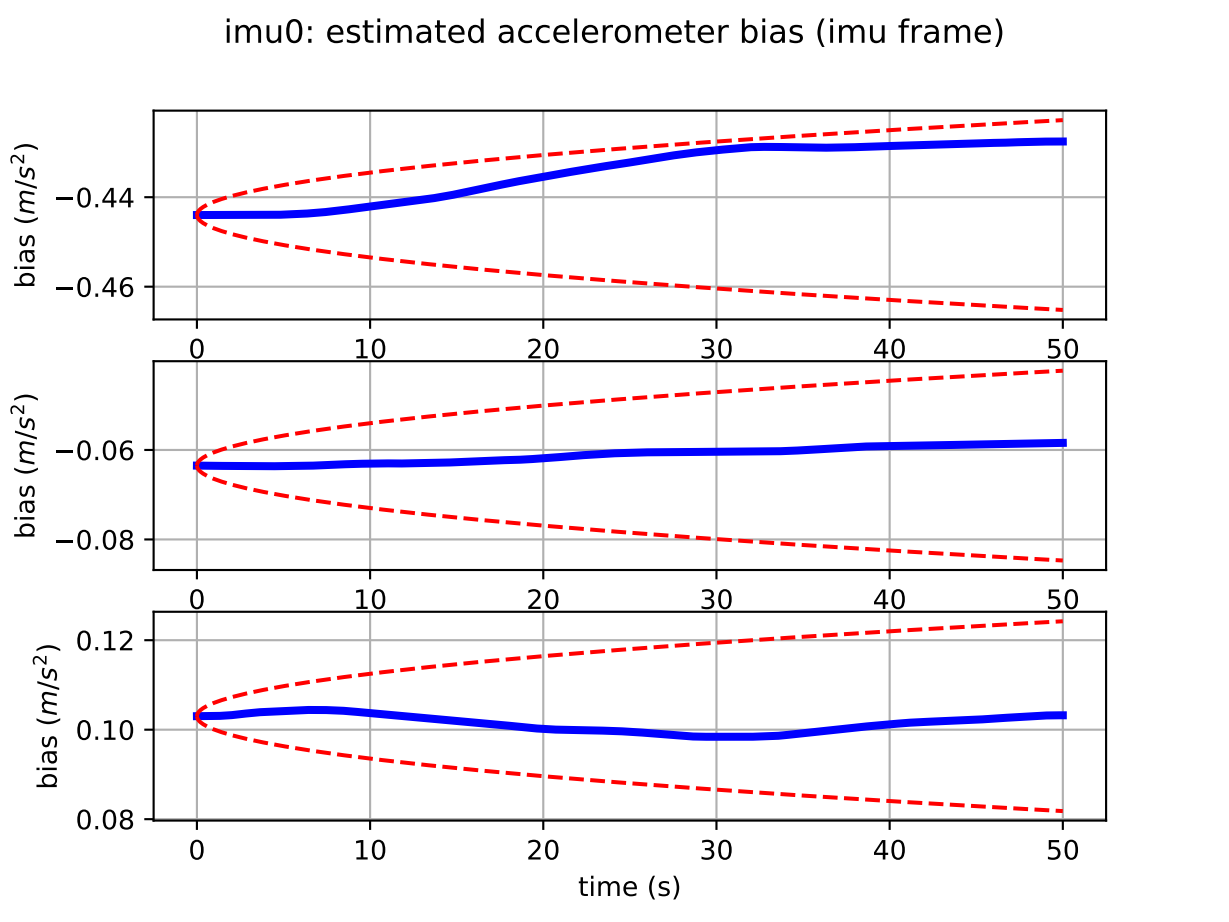
Model: calibrated
Update rate: 400
Accelerometer:
Noise density: 0.02
Noise density (discrete): 0.39999999999999997
Random walk: 0.001
Gyroscope:
Noise density: 0.0005
Noise density (discrete): 0.01
Random walk: 0.0001
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

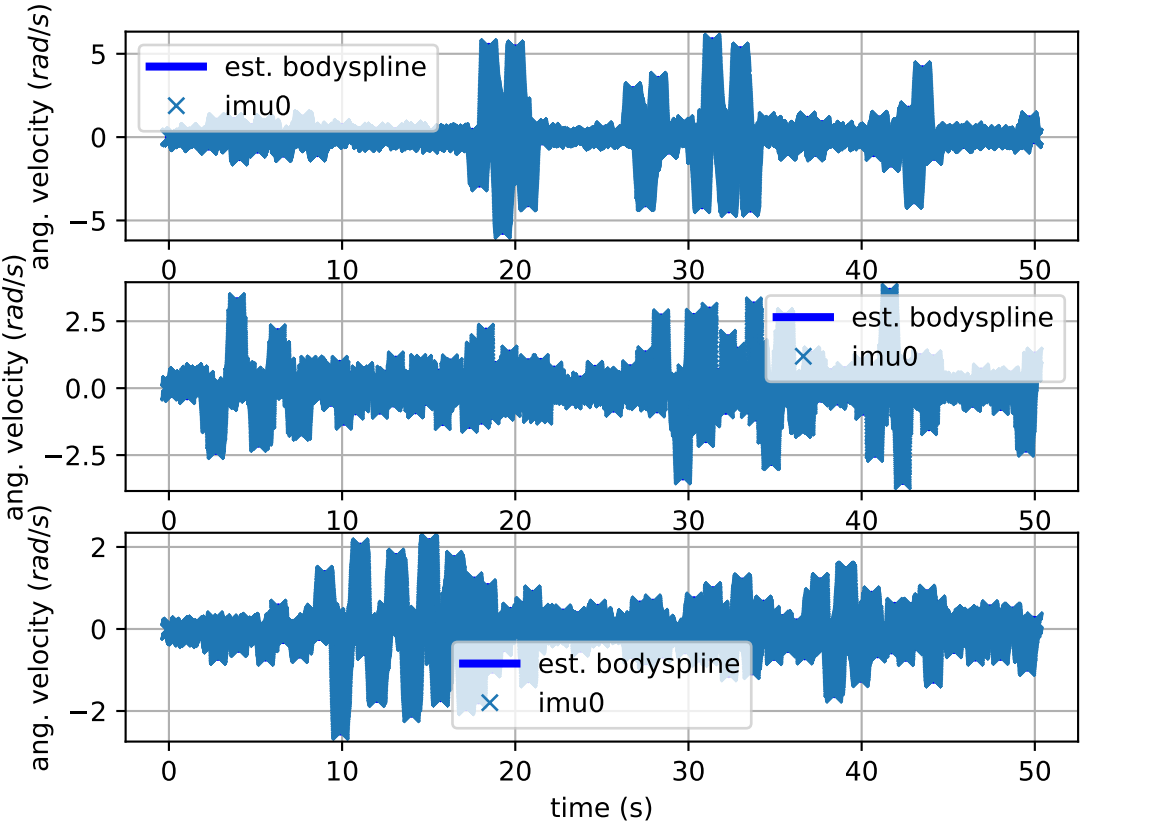

imu0: estimated poses
0.7
0.6
0.5
0.4
0.3
0.2
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4

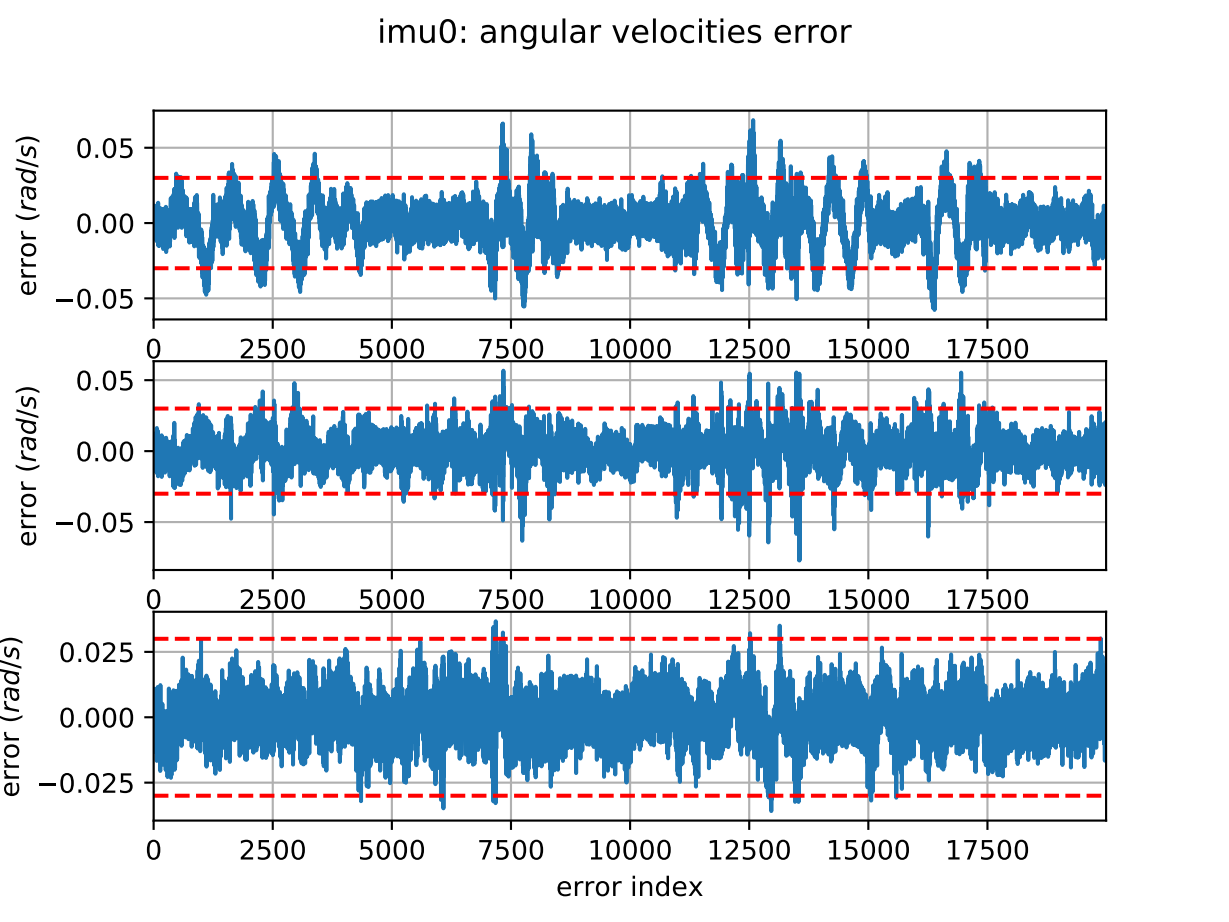

imu0: sample inertial rate
avg dt (ms) = 2.50 +- 0.9819
sample rate (ms)
time (s)
0
2
4
6
8
10
20
30
40

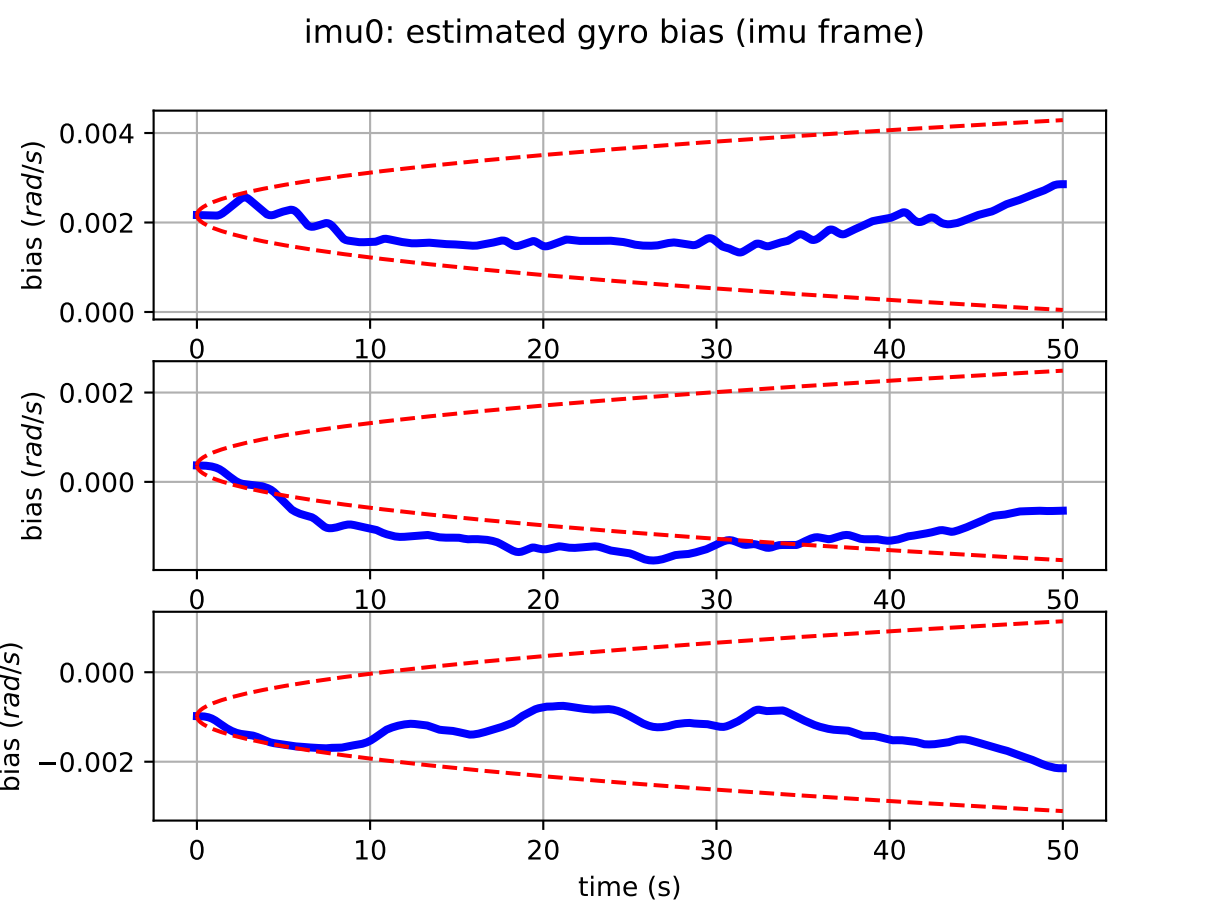

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)

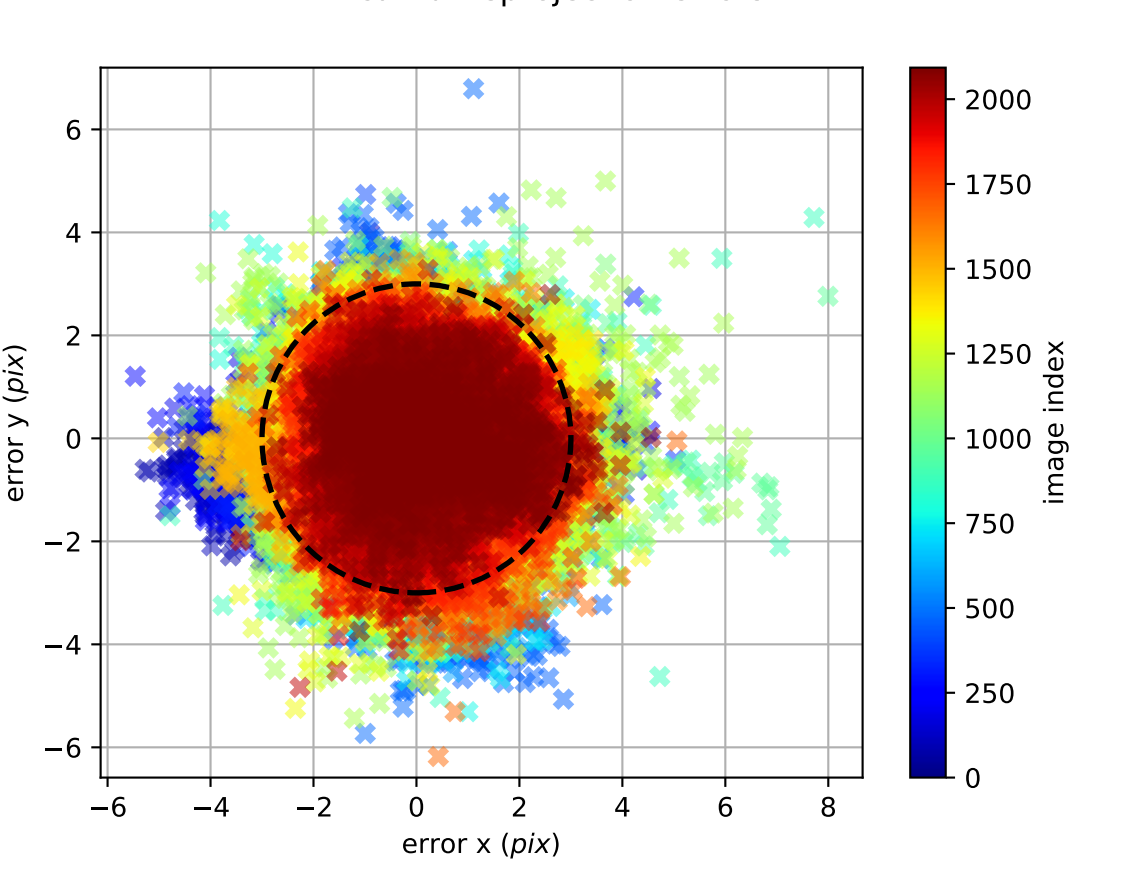

imu0: acceleration error
error (m/s^2)
1
0
−1
0
2500
5000
7500
10000
12500
15000
17500
error index

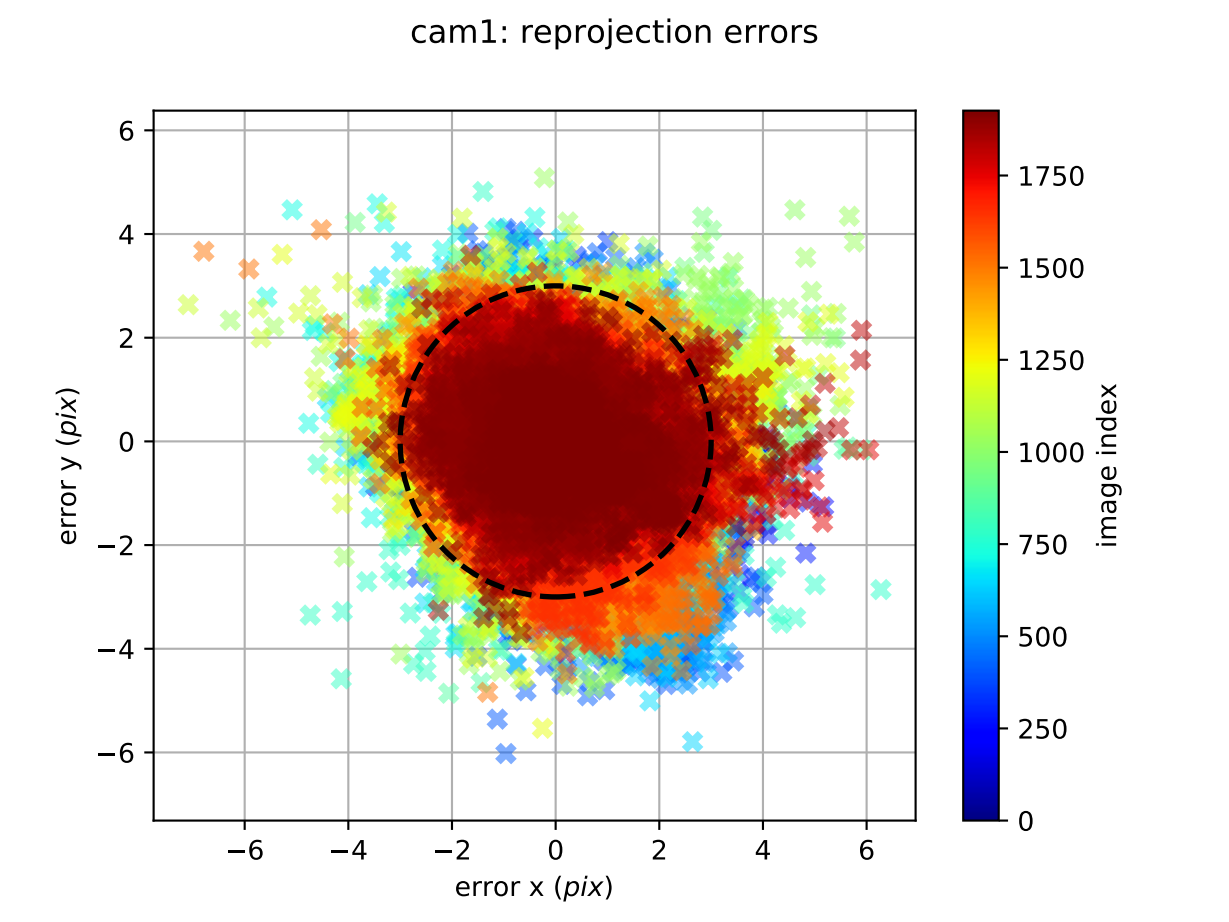

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
−0.44
−0.46
0
10
20
30
40
50
bias (m/s^2)
−0.06
−0.08
0
10
20
30
40
50
bias (m/s^2)
0.12
0.10
0.08
0
10
20
30
40
50
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
5
0
−5
0
10
20
30
40
50
ang. velocity (rad/s)
est. bodyspline
imu0
2.5
0.0
−2.5
0
10
20
30
40
50
ang. velocity (rad/s)
est. bodyspline
imu0
2
0
−2
0
10
20
30
40
50
time (s)


imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
0
2500
5000
7500
10000
12500
15000
17500
error (rad/s)
0.05
0.00
−0.05
0
2500
5000
7500
10000
12500
15000
17500
error (rad/s)
0.025
0.000
−0.025
0
2500
5000
7500
10000
12500
15000
17500
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.004
0.002
0.000
0
10
20
30
40
50
bias (rad/s)
0.002
0.000
bias (rad/s)
0.000
−0.002
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−6
−4
−2
0
2
4
6
8
2000
1750
1500
1250
1000
750
500
250
0


cam1: reprojection errors
error x (pix)
error y (pix)
image index
6
4
2
0
−2
−4
−6
1750
1500
1250
1000
750
500
250
0